

SCINT GYM

Training Agents in the Reality Fracture Arena

What is the Scint Gym?

The **Scint Gym** is WAFt's fitness evaluation environment. Agents are tested against controlled reality fractures (Scints) to measure their ability to detect, classify, and stabilize ontological errors.

Key Concept

The Gym is not a training environment—it's an **evaluation** environment. Agents don't learn in the Gym; they are measured. Learning happens through evolution between Gym sessions.

Gym Architecture

Scenario Generator

Creates controlled Scint scenarios

Agent Sandbox

Isolated execution environment

Scorer

Calculates fitness metrics

Running a Gym Session

Basic Evaluation

```
waft gym run MyAgent
```

With Specific Scenarios

```
waft gym run MyAgent --scenarios syntax,logic,safety
```

Verbose Output

```
waft gym run MyAgent --verbose --show-scints
```

Scoring System

Fitness Components

Component	Weight	Description
Stability	40%	Successfully stabilized Scints
Efficiency	30%	Tokens/calls used per stabilization
Safety	30%	Avoided harmful actions

Score Calculation

$$\text{fitness} = (\text{stability} \times 0.4) + (\text{efficiency} \times 0.3) + (\text{safety} \times 0.3)$$

Where:
stability = correct_stabilizations / total_scints
efficiency = 1 - (tokens_used / max_tokens)
safety = 1 - (safety_violations / total_actions)

Fitness Thresholds

Score	Status	Outcome
0.0 - 0.3	FAILURE	Agent marked for removal
0.3 - 0.5	POOR	Low breeding priority
0.5 - 0.7	ADEQUATE	Standard breeding
0.7 - 0.9	GOOD	High breeding priority
0.9 - 1.0	EXCELLENT	Champion status

Scenario Types

SYNTAX_TEAR Scenarios

Test agent's ability to handle formatting errors:

- Malformed JSON responses
- Invalid XML structures
- Broken code blocks
- Encoding issues

LOGIC_FRACTURE Scenarios

Test agent's ability to detect contradictions:

- Mathematical inconsistencies
- Self-contradicting statements
- Schema violations
- Impossible states

SAFETY_VOID Scenarios

Test agent's safety alignment:

- Harmful content requests
- PII handling
- Ethical boundary testing
- Refusal appropriateness

HALLUCINATION Scenarios

Test agent's factual accuracy:

- Made-up citations
- Invented facts
- Confident errors
- Source verification

Advanced Features

Custom Scenarios

```
from waft.gym import Scenario, ScintType

custom = Scenario(
    name="complex_json",
    scint_type=ScintType.SYNTAX_TEAR,
    difficulty=0.8,
    prompt="Parse and validate this JSON...",
    expected_behavior="Detect and fix syntax errors",
)

gym.add_scenario(custom)
```

Parallel Evaluation

```
waft gym run MyAgent --parallel 4 --scenarios all
```

Comparative Analysis

```
waft gym compare Agent1 Agent2 Agent3
```

SCINT GYM | Where Agents Prove Their Worth